

Axiomatic Probability & Integration

UChicago Political Science Math Camp

Ryan Gibbons & Luke Cain

2026

Roadmap

Probability

Probability is the **likelihood of an event occurring**

We denote the probability of an event A as a real number between 0 and 1:

$$\Pr[A] \in [0, 1]$$

Axioms of Probability

There are three basic axioms when working with probability:

1.

$$\forall a \in A, \Pr[a] \in \mathbb{R}_+ \implies \Pr[a] \geq 0$$

2.

$$\forall A, \sum_{a \in A} \Pr[a] = 1$$

3.

$$a_i \cap a_{-i} = \emptyset \quad \forall a_i, a_{-i} \iff \Pr \left[\bigcup_{i=1}^k a_i \right] = \sum_{i=1}^k \Pr[a_i]$$

Applying Axioms

Show these axioms for each of the following discrete probabilities:

1. A fair two-sided coin
2. An unfair six-sided dice
3. Tomorrow's temperature (in whole degrees, ok to place bounds)

Random Variables

A random variable is a function X such that there exists a real number for every $s \in S$:

$$X : S \rightarrow \mathbb{R}$$

There is then some non-deterministic probability $\Pr[X(s) = x] \in [0, 1]$ for every event s .

PMFs

For a discrete random variable X , we can model a **probability mass function**:

$$\text{PMF} \equiv f(x) = \Pr[X = x]$$

This models the likelihood of any given outcome of $x \in S$.

What do our probability axioms imply about PMFs?

PMF examples

How might we construct the PMFs of the following?

- ▶ A fair, six-sided dice
- ▶ An unfair six-sided dice (twice as likely to land on 6)
- ▶ A nation's global ranking in GDP (assume $n = 200$ for simplicity)

PDFs

We use **probability density functions** (PDFs) to evaluate the “shape” of a continuous random variable:

- ▶ Uniform distribution: $f(x) = \frac{1}{b-a} \iff x \in [a, b]; 0 \text{ else}$

PDFs

We use **probability density functions** (PDFs) to evaluate the “shape” of a continuous random variable:

- ▶ Uniform distribution: $f(x) = \frac{1}{b-a} \iff x \in [a, b]; 0 \text{ else}$
- ▶ Normal distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}$

PDFs

We use **probability density functions** (PDFs) to evaluate the “shape” of a continuous random variable:

- ▶ Uniform distribution: $f(x) = \frac{1}{b-a} \iff x \in [a, b]; 0 \text{ else}$
- ▶ Normal distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}$
- ▶ Exponential distribution: $f(x) = \lambda \exp\{-\lambda x\}$

But, moving from discrete to continuous outcomes presents a mathematical conundrum...

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1?
What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1?
What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **ten-sided** dice. What is the probability of rolling 1?
What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **ten-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **twenty-sided** dice. What is the probability of rolling 1?
What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **ten-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **twenty-sided** dice. What is the probability of rolling 1?
What is the total probability?

Suppose you roll a **one-hundred-sided** dice. What is the probability of rolling 1? What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **ten-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **twenty-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **one-hundred-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **ten-thousand-sided** dice. What is the probability of rolling 1? What is the total probability?

A Simulated Example

Suppose you roll a **four-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **six-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **ten-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **twenty-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **one-hundred-sided** dice. What is the probability of rolling 1? What is the total probability?

Suppose you roll a **ten-thousand-sided** dice. What is the probability of rolling 1? What is the total probability?

Zero-Measure Probability

A fundamental paradox in studying continuous random variables arises: the probability of any specific event is **measure zero**, but the random variable is non-trivial.

This result is *not* tied to the nature of the distribution, other than its continuity.

Given this, how might we study... anything probabilistic?

Cumulative Probability

Suppose we have a uniform random variable $X \sim U[0, 1]$ – that is, it takes on any value between 0 and 1 with equal probability.

Cumulative Probability

Suppose we have a uniform random variable $X \sim U[0, 1]$ – that is, it takes on any value between 0 and 1 with equal probability.

The probability of drawing $x = 0.5$ is measure-zero. But, the probability of drawing some number *less than* 0.5 is well-determined (and is in fact 0.5).

Cumulative Probability

Suppose we have a uniform random variable $X \sim U[0, 1]$ – that is, it takes on any value between 0 and 1 with equal probability.

The probability of drawing $x = 0.5$ is measure-zero. But, the probability of drawing some number *less than* 0.5 is well-determined (and is in fact 0.5).

While probability distributions can show relative likelihood, cumulative metrics can return *exact probabilities*.

CDFs

We often want to represent the probability that a random variable is larger or smaller than some threshold. For this, we can model a **cumulative distribution function**:

$$\text{CDF} \equiv F(x) = \Pr[X \leq x] = \sum_{i \leq x} \Pr[i]$$

What do our probability axioms imply about discrete CDFs? How might we solve $\Pr[X \geq x]$?

CDF examples

How might we construct CDFs of the following?

- ▶ A fair dice rolling at least a 3
- ▶ An individual being in the top n -th percentile of wealth
- ▶ The “amount” of persuasion voters need to choose candidate A in a two-race, majoritarian election

How Could We Do This?

Our previous CDF definition lets us define cumulative probability across discrete outcomes.

But, now we want to define cumulative probability across *continuous* outcomes.

...What if we just made the “buckets” of discrete outcomes infinitely small?



Calculus and Infinity

Calculus fundamentally solves the following contradictions:

- ▶ Derivation solved for the rate of change of a function as $\Delta x \rightarrow 0$, such that $\frac{\text{rise}}{\text{run}} \rightarrow \infty$.
- ▶ Integration solves for the sum of an infinite number of infinitely small segments, where the size of each approaches 0 such that $\sum_i \rightarrow 0$.

In most cases, $\frac{\text{rise}}{\text{run}} \neq \infty$ and $\sum_i \neq 0$.

A Different Approach to Intuition

Say we want to measure the area bounded by some function(s), or the **area beneath the curve**:

1. If all functions are linear, we can use geometric formulas (extensions of $A = bh$) to evaluate
2. If even one function is non-linear¹, then standard geometry can only approximate

¹At least, non-circular, although the area of a circle is derived from calculus anyways

Integration From Approximation

We can approximate the area under a curve by splitting the area into a series of rectangles, such that $A = \sum_i bh_i$

With very large rectangles (i.e., very large b -values), this creates a lot of error. **But what happens** as we add more rectangles, and shrink the size of each?

$$A = \sum_i bh_i \quad \implies \quad b \rightarrow 0, i \rightarrow \infty$$

How does this reflect our derivation definition?

Notation

Integrate $f(x)$ with respect to $x \rightarrow$ What is the sum of every $f(x)$ over every possible x ?

We denote this as

$$\int f(x) dx$$

We call this an **indefinite integral**

A New Limit Definition

$$\int f(x)dx = \lim_{b \rightarrow 0} \sum_i^{\frac{1}{b}} bf(x_i)$$

Integration vs Derivation

- ▶ The integral of $f(x)dx$ represents the sum of all f 's, such that if f represents a rate of change, then $\int f(x)dx$ represents the **change that occurred...**
- ▶ Think: what function's derivative returns $f(x)$? That function represents $\int f(x)dx$

To borrow again from Physics:

$$\text{Jerk} \Rightarrow \int dt = \text{Acceleration} \Rightarrow \int dt = \text{Speed} \Rightarrow \int dt = \text{Position}$$

Definite vs. Indefinite Integrals

Often, we only want to evaluate the integral of a function over some specific domain. To integrate $f(x)$ w.r.t x over the domain $[a, b]$, we write the **definite integral**

$$\int_a^b f(x) dx$$

An indefinite integral often produces a function. A definite integral almost always returns a number (or function independent of x).

$$\int_a^b f'(x)dx = f(b) - f(a)$$

This yields two truths:

1. The integral is the inverse of the derivative (the integral of a function's derivative is just the function)
2. For some defined domain $[a, b]$, the value of the integral over that domain is $\int f(b)dx - \int f(a)dx$

FTOC and Indefinite Integrals

Let's reiterate an important point here:

If we denote the indefinite integral $\int f(x)dx = F$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Then, to solve $\int_a^b f(x)dx$, we evaluate:

$$\int f(x)dx = F(x) \quad \implies \quad \int_a^b f(x)dx = F(x) \Big|_{x=a}^{x=b}$$

How Do We Actually Do This?

The Good Stuff

We can use a series of rules to integrate a function in its general form
for any value of x :

$$f(x) \implies [\text{RULE}] \implies \int f(x) dx$$

Note that now, our indefinite integral is itself another function of x .

The Sum Rule

For all integrals:

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

For definite integrals such that $a \leq b \leq c$:

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

The Power Rule, Backwards

Recall the derivative power rule:

$$\frac{d}{dx}x^k = k \cdot x^{k-1}$$

Then, if $\int f'(x)dx = f(x)$, what function would, when derived, produce ours?

The Power Rule, Backwards

Recall the derivative power rule:

$$\frac{d}{dx}x^k = k \cdot x^{k-1}$$

Then, if $\int f'(x)dx = f(x)$, what function would, when derived, produce ours?

$$\int x^k dx = \frac{x^{k+1}}{k+1}$$

plus one little thing...

Remember Our Constants?

1. We know that the derivative of a constant is 0, such that $\frac{d}{dx}c = 0 \forall c \in \mathbb{R}$
2. If integration “reverts” a derivative, then it must account for any constant terms zero-d out in derivation
3. We resolve this as follows:

$$\int x^k dx = \frac{x^{k+1}}{k+1} + c$$

even though we don't know what c actually is.

A Power Rule Corollary (Reversed)

1. We know that $\int x^k dx = \frac{1}{k+1} x^{k+1}$ (**Power Rule**) and that $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$ (**Sum Rule**)
2. We also know that for a constant ℓ , $\ell \cdot n = \sum_1^\ell n$, or n added ℓ times.
3. Then, the integral is the sum of x^k added up ℓ times:

$$\int \left[\ell \cdot x^k \right] dx = \sum_1^\ell \int x^k dx$$

4. We know that $\sum_1^n x = nx$, so

$$\sum_1^\ell \int x^k dx = \ell \cdot \int x^k dx \quad \Rightarrow \quad \int \left[\ell \cdot x^k \right] dx = \ell \cdot \frac{1}{k+1} \cdot x^{k+1}$$

$\Rightarrow$ **Coefficients:** The coefficient of a term is kept as a coefficient of that term's integral (i.e., multiply $\frac{1}{k+1}$ by the coefficient as well)

Quick Note on Constants

Recall that a constant m can be represented as $m \cdot x^0$. Then,

$$\begin{aligned}\int m dx &= \int m \cdot x^0 dx \\ \implies m \cdot \int x^0 dx \\ \implies m \cdot \frac{x^{0+1}}{0+1} + c &= m \cdot \frac{x^1}{1} + c \\ \therefore \int m dx &= mx + c\end{aligned}$$

Some Examples

Let's try some simple examples:

$$\int x^2 + 2x^3 dx$$

Some Examples

Let's try some simple examples:

$$\int x^2 + 2x^3 dx$$

$$\int 2x^3 + \frac{1}{2}x^4 dx$$

Some Examples

Let's try some simple examples:

$$\int x^2 + 2x^3 dx$$

$$\int 2x^3 + \frac{1}{2}x^4 dx$$

$$\int 3 + \sqrt[3]{x} dx$$

Some Examples

Let's try some simple examples:

$$\int x^2 + 2x^3 dx$$

$$\int 2x^3 + \frac{1}{2}x^4 dx$$

$$\int 3 + \sqrt[3]{x} dx$$

$$\int x^{-3} + x^{-2} + x^{-1} dx$$

Closing Some Subtle Foreshadowing

The power rule doesn't work for $f(x) = x^{-1} = \frac{1}{x}$, as it solves:

$$\int x^{-1} dx = \frac{1}{-1+1} x^{-1+1} = -\frac{x^0}{0}$$

Thankfully, our natural logarithm solves this for us:

$$\int \frac{1}{x} dx = \ln(x) + c$$

Integrating Exponentials

We have a specific rule for integrating exponentials and logarithms:

$$\int a^x dx = \frac{a^x}{\ln a} + c$$

$$\int \log_b x dx = \frac{x}{\ln b} (\ln b - 1) + c$$

Definite Integration

So far, we've expressed $F(x)$ for every $f(x)$, both as functions of x , as *indefinite integrals*. When we add bounds of integration, we simply use FTOC:

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int_2^4 x^3 dx = \left. \int x^3 dx \right|_2^4$$

$$\Rightarrow \left. \frac{x^4}{4} + c \right|_2^4$$

But, we don't know c ...

c you later

There's a handy trick with definite integrals:

$$\begin{aligned}\int f(x)dx &= F(x) + c \\ \implies \int_a^b f(x)dx &= F(b) + c - (F(a) + c) \\ \implies \int_a^b f(x)dx &= F(b) - F(a)\end{aligned}$$

For definite integrals, we subtract $c - c$ and can therefore exclude it. So,

$$\left. \frac{x^4}{4} + c \right|_2^4 = \frac{4^4}{4} - \frac{2^4}{4} = 16$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

$$\int_2^6 \frac{1}{2}x^2 + 4 dx$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

$$\int_2^6 \frac{1}{2}x^2 + 4 dx$$

$$\int_0^b \sqrt{x} + \sqrt{2} dx$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

$$\int_2^6 \frac{1}{2}x^2 + 4dx$$

$$\int_0^b \sqrt{x} + \sqrt{2}dx$$

$$\int_a^2 \frac{1}{x^2} dx$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

$$\int_2^6 \frac{1}{2}x^2 + 4dx$$

$$\int_0^b \sqrt{x} + \sqrt{2}dx$$

$$\int_a^2 \frac{1}{x^2} dx$$

A Turn For The Worst

Two of our fundamental derivative rules – the **chain rule** and the **product rule** – are a bit more complicated to invert:

$$\frac{d}{dx}f(u) = f'(u) \cdot u'$$

$$\implies F(u) = \int f(u) \cdot u du \dots$$

There's not a clear solution here

U-Substitution

The inverse of the Chain Rule is called **integration by substitution**, or “**u-sub**”, and works as follows:

1. Express the integral in the form

$$\int f(g(x))g'(x)dx$$

2. Assign $g(x) = u$ such that the integral can be rewritten:

$$\int f(u)du$$

3. Evaluate the integral:

$$\int f(u)du = F(u)$$

4. Substitute back $u = g(x)$:

$$\int f(g(x))g'(x)dx = F(g(x))$$

Integration By Parts

Integration by parts inverts the product rule. We will not cover it in math camp, but you should be aware of the concept and ready to learn its process.

Improper Integrals

For definite integrals which do not converge to $f(x) = a$, or for any asymmetric integrals whereby $f(x) = \pm\infty$ for $x \in \mathbb{R}$, we call that integral **improper**.

This can either result in a finite value, such as $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$, or an undefined value, such as $\int_{-\infty}^{\infty} \frac{1}{x} dx$

Gabriel's Horn

Imagine a horn formed by the function $f = \frac{1}{x}$ for the domain $x \in [1, \infty)$

What is its volume? What is its surface area?

Horn Volume

We know that $V = a \cdot h$; in this case we can set $h = \infty$ and $a = \pi r(x)^2 = \frac{\pi}{x^2}$. The volume is then the sum of this area over every possible height:

$$V = \int_1^{\infty} \frac{\pi}{x^2} dx = -\frac{\pi}{x} \Big|_1^{\infty}$$

$$V = \pi$$

The volume of the horn is finite.

Horn Surface Area

We know that surface area can be calculated as $s = c \cdot h$. In this case, we set $h = \infty$ and $c = \pi 2r = \frac{2\pi}{x}$. The surface area is then the sum of this circumference over every possible diameter:

$$s = \int_1^{\infty} \frac{2\pi}{x} dx = 2\pi \cdot \ln(x) \Big|_1^{\infty}$$

$$s = \infty$$

The horn has finite volume but infinite surface area. *You could fill the inside with paint, but never fully paint the outside.*

Expectation

You've already seen the expectation operator for discrete random variables:

$$\mathbb{E}[X] = \sum_{x \in X} x \cdot \Pr[X = x]$$

How might we apply this to continuous random variables?

Expectation

You've already seen the expectation operator for discrete random variables:

$$\mathbb{E}[X] = \sum_{x \in X} x \cdot \Pr[X = x]$$

How might we apply this to continuous random variables?

Send $\#\{X\} \rightarrow \infty$ and $\Pr[X = x] \rightarrow \frac{1}{\infty}$.

Expectation With Infinite Outcomes

If we're just taking an infinite sum of infinitely small pieces, we're taking an integral.

This yields our continuous expectation function:

$$\mathbb{E}[X] = \int_{x \in X} xf(X)dx$$

where $f(x)$ is the PDF.

Example

Suppose we have an exponential distribution where $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$:

$$f(x) = \lambda \exp\{-\lambda x\}$$

Example

Suppose we have an exponential distribution where $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$:

$$f(x) = \lambda \exp\{-\lambda x\}$$

We can integrate accordingly:

$$\begin{aligned}\mathbb{E}[f(x)] &= \int x f(x) dx \\ &= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\ &= \frac{1}{\lambda}\end{aligned}$$

Expected Utility

Often, we have two functions (i.e., a PDF and a utility function), and we want to know the expectation of their product.

We can use a relatively simple integral:

$$\mathbb{E}[u(X)] = \int_{\mathbb{R}} u(x)f(x)dx$$

An Example: Risk vs. Reward

Suppose we play some lottery where (1) returns are marginally increasing and (2) higher rewards are less likely than lower rewards:

$$f(x) = \lambda \exp\{-\lambda x\}$$
$$u(x) = \lambda x^a$$

for $a > 1$. What is the expected value of this lottery?

Answer

$$\begin{aligned}\mathbb{E}[u(X)] &= \int_{\mathbb{R}} f(x)u(x)dx \\ &= \int_0^{\infty} \lambda \exp\{-\lambda x\}x^a dx \\ &= a\end{aligned}$$

Let us Optimize

If you were playing the above lottery, how would you manipulate λ , a to maximize your expected utility?